

Assignment 3

Question 1: Fill Table 1(Core Logic Design of Trading Strategies) and Table 2 (Design of Algorithmic Models).

Core Logic Design of Trading Strategies		
Targeting Market	U.S. Equities: Publicly traded companies in commodity-intensive industries (industrials, consumer staples, agriculture, energy)	
Phenomena Description	Theoretical Framework	Phenomena: When a commodity input cost rises significantly, companies that rely on that input face margin compression, yet stock prices frequently do not adjust immediately. This lag, information diffusion lag, creates a pricing inefficiency. Framework: Efficient Market Hypothesis (semi-strong form) would suggest stock prices should immediately incorporate public commodity data. This strategy exploits the documented deviation from EMH, grounded in behavioral finance (investor inattention, analyst forecast stickiness).

Analytical Framework ☐	Theoretical Hypothesis ☐	Framework: Commodity-equity spread analysis. Calculate rolling cost-of-goods-sold (COGS) sensitivity (beta) of each equity to its primary input commodity. Compare realized commodity price change vs. implied stock price adjustment. Hypothesis: A statistically significant increase (or decrease) in a key input commodity price, which is not reflected in the dependent company's stock price within a defined lag window, predicts subsequent downward (upward) price correction in the stock. $Spread_t = (\beta \times \Delta C_t) - \Delta S_t$ Meaning, the Spread of mispricing gap at a given time t, is the difference of the change in stock (S) and product of the stock's sensitivity to the commodity moves (β) and the change in the commodity (C).
Phenomena Interpretation ☐	Theoretical Reasoning ☐	Interpretation: The lag represents investor inattention to cross-asset signals, analyst earnings estimate stickiness, and institutional delay in repricing. Equity markets are slower to process commodity market signals than commodity-forward markets. Reasoning: Companies cannot indefinitely absorb input cost increases without earnings erosion, nor will they forgo profit expansion when input costs fall. Fundamental value must eventually converge, creating a predictable directional trade.

Trading Opportunity	Cross-asset pricing gap between commodity futures/spot prices and the stock prices of companies with high commodity input dependency. Gap closes when analyst revisions, earnings surprises, or market repricing occurs (expected within 2–8 weeks of a significant commodity move).
Trading Operations	Long or short equity positions on identified stocks when commodity-equity spread exceeds a statistical threshold (relative to standard deviation). Enter on signal trigger; exit at mean-reversion target or stop-loss. No direct commodity position taken, rather this is purely equity play on input-cost mispricing.
Financial Instrument(s)	U.S.-listed equities (common stock).

Design of Algorithmic Models	
Data Requirement	Daily commodity spot/futures prices (aluminum LME, corn CBOT, natural gas NYMEX, crude oil WTI, copper LME, wheat CBOT). Daily OHLCV equity data for target companies. Company COGS and gross margin data from quarterly 10-Q filings. Optional: analyst EPS estimates for divergence scoring.
Quantitative Model	1. Sensitivity Estimation: Regress historical stock returns on commodity price changes to derive commodity-beta (β) for each stock-commodity pair over a rolling 252-day window. 2. Expected vs. Realized Price: Compute "implied stock move" = $\beta \times \text{commodity \% change over the signal window}$

	(e.g., 10-day). Compute "actual stock move" over same window. 3. Spread Signal: $\text{Spread} = \text{Implied Move} - \text{Actual Move}$. If $\text{Spread} > \text{threshold}$ (e.g., 1.5σ of historical spread), a signal is generated. 4. Engle-Granger Test: This tests the cointegration of pairs, evaluating if a pair has a statistically significant mean-reverting relationship. Further, the Granger causality will be used to test if the commodity price movements leads the equity price movement.
Strategy Type	Statistical Arbitrage (Cross-Asset Mean Reversion). Specifically: Commodity-Equity Input Cost Pass-Through Arbitrage. Secondary elements of event-driven (earnings reaction) and momentum (commodity trend confirmation).
Trading Decision Triggers	Threshold-based trigger: $\text{Spread} > +1.5\sigma \rightarrow \text{Short equity}$ (stock overpriced relative to rising input cost). $\text{Spread} < -1.5\sigma \rightarrow \text{Long equity}$ (stock underpriced relative to falling input cost). Exit trigger: Spread reverts within 0.25σ of zero, OR maximum holding period (e.g., 20 trading days) reached, OR stop-loss at -2% position NAV. (I will not be implementing stop-loss immediately) Note: 1.5, 0.25, -2 are initial estimates subject to revision following backtesting and statistical analysis.
Expected Return Level (1-5)	3, Moderate. Returns driven by mean-reversion episodes; alpha is consistent but per-trade returns are modest (1–4% target per trade).

Expected Risk Level (1-5)	2, Low-Moderate. Main risks: divergence persistence (commodity-equity spread widens further before reverting), macro regime shifts, and company-specific news unrelated to input costs.
Expected Capacity Level (1-5)	3, Medium. Strategy is applied to large/mid-cap equities with sufficient liquidity. Capacity constrained by number of signal opportunities across the defined stock universe (~20-50 pairs).
Expected Frequency Level (1-5)	2, Low-Moderate. Signals generated weekly to monthly depending on commodity volatility; not high-frequency. Average holding period: 5–90 trading days.
Parameters* <i>(definition and possible ranges)</i>	β_sensitivity (commodity-beta): Rolling OLS regression window. Range: 60–252 trading days. signal_window: Lookback period over which commodity and equity moves are measured. Range: 5–90 trading days. entry_threshold (σ): Z-score of spread required to trigger entry. Range: $1.0\sigma - 2.5\sigma$. exit_threshold (σ): Spread reversion level to close position. Range: $0.0\sigma - 0.5\sigma$. max_holding_days: Maximum days before forced exit. Range: 10–30 trading days. stop_loss: Position-level loss limit. Range: –1.5% to –3.0% of position NAV. (future)

Comodity-Equity Pair Examples

1. Natural Gas - Fertilizer Producers (CF Industries, Mosaic, Nutrien)

Natural gas is ~70-80% of ammonia production cost. High, direct, quantifiable input relationship (strong academic backing).

2. Copper - Electric Vehicle/EV Component Manufacturers (e.g., TE Connectivity, Aptiv)

Copper is a critical input for wiring harnesses, motors, and batteries. Highly newsworthy, liquid commodity with growing equity sensitivity.

3. Crude Oil - Airline Stocks (Delta, United, Southwest)

Jet fuel (~25-30% of airline operating costs) is directly derived from crude. Well-studied relationship, large price moves create clear signals, high analytical tractability.

4. Steel (HRC Futures) - Auto Manufacturers (Ford, GM, Stellantis)

Steel is a primary input for vehicle bodies; moves in hot-rolled coil prices directly affect COGS. Strong margin sensitivity and easy-to-model relationship.

Question 2: Quant Strategy Strength.

This strategy is strong because it is grounded in a core phenomenon, rigorously and quantitatively selects pairs and uses statistical methodology to define parameters.

1. Core Phenomenon. The core phenomenon is grounded in real market behavior. Information diffusion lag does not rely on broken markets or irregularities, but rather a natural phenomenon that exists in markets. The semi-strong EMH deviation exploits a bounded inefficiency, but does not claim directly that markets are wrong. This is based in behavioral finance mechanisms where investor inattention and analyst forecast stickiness

are well-documented phenomena in academic literature. This gives my lag hypothesis a theoretical foundation beyond just intuition

2. Rigorous Commodity Pair Selection. The initial set of pairs are ones where the commodity is a dominant, not merely notable, cost driver. Further, the statistical methodology used to evaluate and select pairs distinctively separates this quantitative model. By grounding in Engle-Granger Cointegration, I confirm that the pair has a stationary, mean-reverting spread, rather than simply trading on correlations. Further, the Granger causality test is another mathematical solution to test the directional hypothesis directly. With these tests, I trade pairs where the data confirms both that a stable relationship exists and that the direction of information flow follows my hypothesis.
3. Parameter Definitions. In all parameter definitions, I hold that I start with initial estimates that are subject to revision following backtesting and statistical analysis. This allows the data to determine the exact trading principles. I allow for ranges for my Beta, entry threshold and max holding period. By using an adaptive Beta, the model continuously updates the stock's sensitivity to its commodity. Finally by using spread, I am not simply comparing raw commodity and equity prices, but rather measuring the gap between what the stock should have moved and what it actually moved.

Question 3: Clear and Specific Trading Strategy/Rules.

Step 1: Validate the pair using the Engle-Granger cointegration test. P-value must be < 0.05 .

Then run the Granger causality test to ensure commodity leads equity price change.

Step 2: Estimate Beta, Spread and Z-score daily.

- Beta (β): estimated via rolling 252-day OLS regression of daily returns
- Spread = $(\beta \times \% \text{ change in commodity}) - (\% \text{ change in equity})$
- $Z = (\text{Spread} - 60\text{-day rolling mean of Spread}) / (60\text{-day rolling std of Spread})$

Entry Rules:

- Commodity move has persisted for ≥ 3 consecutive days in the same direction
- Not within 5 trading days of the equity's earnings announcement
- VIX is below 30

When Z score:

$Z > +1.5\sigma \rightarrow$ Short the equity (commodity rose, stock hasn't fallen yet)

$Z < -1.5\sigma \rightarrow$ Long the equity (commodity fell, stock hasn't risen yet)

Size each position at maximum 5–10% of total capital

Exit on whichever triggers first:

- Z-score returns to $\leq 0.25\sigma \rightarrow$ Exit, take profit (spread has closed)
- 20 trading days elapsed since entry $\rightarrow$ Exit regardless (time-based close)

AI Acknowledgement: I used Claude 4.6 to better understand what methodology would be best and to come up with a statistically sound trading strategy. All questions and answers can be found in this chat: <https://claude.ai/chat/7e1d6cd3-ddba-48a0-9c79-286fd770d2cc>